

MATH 350-2 Advanced Calculus

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Outline

- 1 Real Analysis Lecture 14
 - Complete metric spaces
 - Limit of a function
 - Continuity

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Complete spaces

Theorem

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If the set X of all values of $\{x_n\}$ is finite, then the Cauchy condition implies $\{x_n\}$ converges.

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TIME OUT!



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To see this, suppose otherwise.

Then for all $x \in S$, there exists $r_x > 0$ such that $B_S(x, r_x) \cap X$ is empty if $x \notin X$ or equal to $\{x\}$ if $x \in X$.

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The open cover $\{U_i : i \in I\}$ with $I = S$ and $U_i = B_S(i; r_i)$ of S has no finite subcover.

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This is a contradiction, so S has an accumulation point of X . □

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Then $\exists N > 0$ such that $m, n \geq N$ implies $d(x_m, x_n) < \epsilon/2$.

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Moreover, $B_S(L, \epsilon/2)$ contains infinitely many points of X .

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It follows that for all $n \geq N$,

$$d(x_n, L) \leq d(x_n, x_m) + d(x_m, L) < \epsilon/2 + \epsilon/2 = \epsilon.$$

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It follows that for all $n \geq N$,

$$d(x_n, L) \leq d(x_n, x_m) + d(x_m, L) < \epsilon/2 + \epsilon/2 = \epsilon.$$

Since $\epsilon > 0$ was arbitrary, this proves $\{x_n\}$ converges to L . □

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Rigorous: For all $\epsilon > 0$ there exists $\delta > 0$ such that

$$0 < |x - a| < \delta \Rightarrow |f(x) - L| < \epsilon.$$

Limit of a function

Let (S, d_S) and (T, d_T) be metric spaces and $A \subseteq S$.

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Then for any accumulation point p of A and element $L \in T$, we say that the **limit as x approaches p of $f(x)$ is L**

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$$x \in B_S(p; \delta) \setminus \{p\} \Rightarrow f(x) \in B_T(L; \epsilon).$$

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